MATHS MANTHAN

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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 01
Chapter 11: Areas Related to Circles

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE

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Q1. If the circumference and the area of a circle are numerically equal, then the radius of the circle is: [1]

(A) 1 unit (B) 2 units (C) π units (D) 4 units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Equating Circumference and Area [0.5 Mark]

Given numerically: Circumference = Area $\implies 2\pi r = \pi r^2$.

Step 2: Solving for Radius r [0.5 Mark]

$\pi r(r-2) = 0$. Since radius must be positive $(r > 0)$, we divide by $\pi r \neq 0$ to get:

$$r = 2 \text{ units}$$

Correct Option: (B) 2 units

Radius of the circle $r = \mathbf{2}$ units

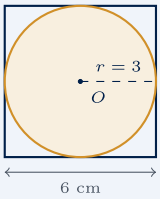
Q2. The area of the circle that can be inscribed in a square of side 6 cm is: [1]

(A) 36π cm^2 (B) 18π cm^2 (C) 12π cm^2 (D) 9π cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Relation between Side and Inscribed Circle [0.5 Mark]

The diameter of a circle inscribed in a square equals the side length of the square:



$$\text{Diameter } d = \text{Side of square } a = 6 \text{ cm} \implies r = \frac{6}{2} = 3 \text{ cm}$$

Step 2: Area Calculation [0.5 Mark]

$$\text{Area} = \pi r^2 = \pi \times (3)^2 = 9\pi \text{ cm}^2$$

Correct Option: (D) 9π cm^2

Area of inscribed circle $= \mathbf{9\pi}$ cm$^\mathbf{2}$

Q3. If the area of a circle is equal to the sum of the areas of two circles of diameters 10 cm and 24 cm, then the diameter of the larger circle is: [1]

(A) 34 cm (B) 26 cm (C) 17 cm (D) 14 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Radii of the Two Circles [0.5 Mark]

$r_1 = \frac{10}{2} = 5$ cm and $r_2 = \frac{24}{2} = 12$ cm.

Let R be the radius of the larger circle.

$$\pi R^2 = \pi r_1^2 + \pi r_2^2 \implies R^2 = r_1^2 + r_2^2 = 5^2 + 12^2 = 25 + 144 = 169$$

Step 2: Diameter of the Larger Circle [0.5 Mark]

$$R = \sqrt{169} = 13 \text{ cm} \implies \text{Diameter } D = 2R = 2 \times 13 = 26 \text{ cm}$$

Correct Option: **(B) 26 cm**

Diameter of larger circle $D = \mathbf{26}$ cm

Q4. The perimeter of a semicircular protractor of radius r is given by: [1]

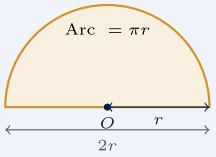
(A) πr (B) $\pi r + r$ (C) $\pi r + 2r$ (D) $2\pi r + 2r$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Formula for Semicircular Boundary [0.5 Mark]

A protractor is closed by both a curved semicircular arc and a straight diameter:

Perimeter = Length of semicircular arc + Diameter



Step 2: Algebraic Expression [0.5 Mark]

$$\text{Perimeter} = \frac{1}{2}(2\pi r) + 2r = \pi r + 2r = (\pi + 2)r$$

Correct Option: **(C)** $\pi r + 2r$

Perimeter of protractor $= \pi\mathbf{r} + \mathbf{2r}$

Q5. **Assertion (A):** If the areas of two circles are in the ratio 9 : 16, then their circumferences are in the ratio 3 : 4.

Reason (R): If the areas of two circles are A_1 and A_2, then the ratio of their circumferences is $\sqrt{A_1} : \sqrt{A_2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Verifying Assertion (A) [0.5 Mark]

Let the radii be r_1 and r_2.

$$\frac{A_1}{A_2} = \frac{\pi r_1^2}{\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \frac{9}{16} \implies \frac{r_1}{r_2} = \sqrt{\frac{9}{16}} = \frac{3}{4}$$

Ratio of circumferences $= \frac{2\pi r_1}{2\pi r_2} = \frac{r_1}{r_2} = \frac{3}{4} = 3:4$. Thus, Assertion (A) is **True**.

Step 2: Verifying Reason (R) [0.5 Mark]

Ratio of circumferences $= \frac{C_1}{C_2} = \frac{2\pi r_1}{2\pi r_2} = \frac{r_1}{r_2} = \sqrt{\frac{\pi r_1^2}{\pi r_2^2}} = \frac{\sqrt{A_1}}{\sqrt{A_2}}$.

Thus, Reason (R) is **True** and provides the exact mathematical rationale for (A).

Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Correct Option: (**A**)

SECTION B: Very Short Answer Type

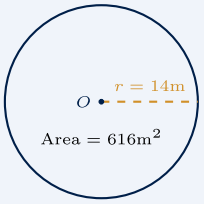
MATHS MANTHAN | **(1 × 2 = 2 Marks)**

Q6. The area of a circular playfield is 616 m^2. Find its circumference. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Find the Radius from Given Area [1 Mark]

Let the radius of the playfield be r m.



$$\text{Area} = \pi r^2 = 616 \text{ m}^2$$

$$\frac{22}{7} r^2 = 616 \implies r^2 = \frac{616 \times 7}{22} = 28 \times 7 = 196$$

$$r = \sqrt{196} = 14 \text{ m}$$

Step 2: Calculate the Circumference [1 Mark]

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 14 = 2 \times 22 \times 2 = 88 \text{ m}$$

Circumference of the playfield = **88** m

SECTION C: Short Answer Type

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q7. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find:

(i) the length of the arc,

(ii) the area of the sector formed by the arc, and

(iii) the area of the segment formed by the corresponding chord. $\left(\text{Use } \pi = \frac{22}{7}, \sqrt{3} = 1.732\right)$

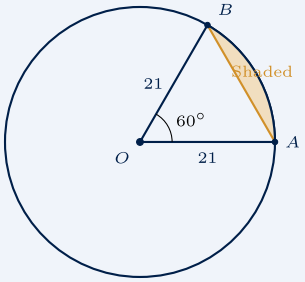
[3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Radius $r = 21$ cm, Central Angle $\theta = 60°$.

Part (i): Length of the Arc (l) [1 Mark]

$$l = \frac{\theta}{360°} \times 2\pi r = \frac{60°}{360°} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22 \text{ cm}$$



Part (ii): Area of the Sector [1 Mark]

$$\text{Area} = \frac{\theta}{360°} \times \pi r^2 = \frac{60°}{360°} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{6} \times 1386 = 231 \text{ cm}^2$$

Part (iii): Area of the Corresponding Segment [1 Mark]

In $\triangle OAB$, $OA = OB = 21$ cm and $\angle AOB = 60° \implies \triangle OAB$ is equilateral.

$$\text{Area of } \triangle OAB = \frac{\sqrt{3}}{4} r^2 = \frac{\sqrt{3}}{4} \times (21)^2 = \frac{441\sqrt{3}}{4} \text{ cm}^2$$

$$= \frac{441 \times 1.732}{4} = \frac{763.812}{4} = 190.953 \text{ cm}^2 \approx 190.95 \text{ cm}^2$$

$$\text{Area of segment} = \text{Area of sector} - \text{Area of } \triangle OAB$$

$$= 231 - 190.95 = 40.05 \text{ cm}^2 \quad \left[\text{or } \left(231 - \frac{441\sqrt{3}}{4}\right) \text{cm}^2\right]$$

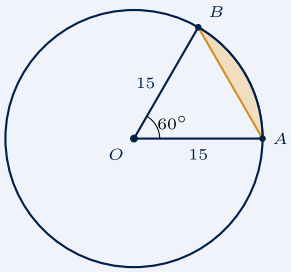
(i) $l = \mathbf{22}$ cm, (ii) Area of Sector = $\mathbf{231}$ cm^2, (iii) Area of Segment = $\mathbf{40.05}$ cm^2

Q8. A chord of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments of the circle. (Use $\pi = 3.14, \sqrt{3} = 1.73$) **[3]**

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► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Radius $r = 15$ cm, Central Angle $\theta = 60°$.



Step 1: Area of Minor Sector & $\triangle OAB$ [1 Mark]

$$\text{Area of minor sector} = \frac{60°}{360°} \times \pi r^2 = \frac{1}{6} \times 3.14 \times (15)^2 = \frac{706.5}{6} = 117.75 \text{ cm}^2$$

Since $\angle AOB = 60°$ and $OA = OB$, $\triangle OAB$ is an equilateral triangle:

$$\text{Area of } \triangle OAB = \frac{\sqrt{3}}{4} r^2 = \frac{1.73}{4} \times 225 = 97.3125 \text{ cm}^2$$

Step 2: Area of Minor Segment [1 Mark]

$$\text{Area of minor segment} = \text{Area of sector} - \text{Area of } \triangle OAB$$
$$= 117.75 - 97.3125 = 20.4375 \text{ cm}^2 \approx 20.44 \text{ cm}^2$$

Step 3: Area of Major Segment [1 Mark]

$$\text{Total Area of Circle} = \pi r^2 = 3.14 \times (15)^2 = 706.5 \text{ cm}^2$$
$$\text{Area of major segment} = \text{Area of circle} - \text{Area of minor segment}$$
$$= 706.5 - 20.4375 = 686.0625 \text{ cm}^2 \approx 686.06 \text{ cm}^2$$

Area of Minor Segment = **20.44** cm^2, Area of Major Segment = **686.06** cm^2

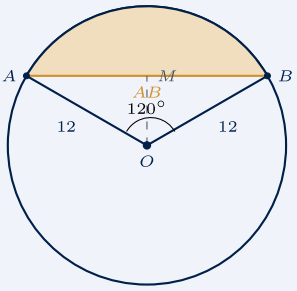
Q9. A chord of a circle of radius 12 cm subtends an angle of 120° at the centre. Find the area of the corresponding minor segment of the circle. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) [3]

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▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: Radius $r = 12$ cm, Central Angle $\theta = 120°$.

Step 1: Area of Minor Sector [1 Mark]



$$\text{Area of minor sector} = \frac{120°}{360°} \times \pi r^2 = \frac{1}{3} \times 3.14 \times (12)^2 = 48 \times 3.14 = 150.72 \text{ cm}^2$$

Step 2: Area of $\triangle OAB$ [1 Mark]

Draw $OM \perp AB$. In isosceles $\triangle OAB$, OM bisects $\angle AOB$ and chord AB:

$$\angle AOM = \angle BOM = 60°$$

In right $\triangle OMA$:

$$OM = OA \cos 60° = 12 \times \frac{1}{2} = 6 \text{ cm}$$

$$AM = OA \sin 60° = 12 \times \frac{\sqrt{3}}{2} = 6\sqrt{3} \text{ cm} \implies AB = 2AM = 12\sqrt{3} \text{ cm}$$

$$\text{Area of } \triangle OAB = \frac{1}{2} \times AB \times OM = \frac{1}{2} \times 12\sqrt{3} \times 6 = 36\sqrt{3} \text{ cm}^2$$

$$= 36 \times 1.73 = 62.28 \text{ cm}^2$$

Step 3: Area of Minor Segment [1 Mark]

$$\text{Area of segment} = \text{Area of sector} - \text{Area of } \triangle OAB$$

$$= 150.72 - 62.28 = 88.44 \text{ cm}^2$$

$$\boxed{\text{Area of Minor Segment} = \mathbf{88.44} \text{ cm}^2}$$

SECTION D: Case-Based / Applied Situation

MATHS MANTHAN | (1 × 4 = 4 Marks)

Q10. Case Study: Round Dining Table Cover with Hexagonal Border Designs

A craftsman creates an embroidered circular table cover for a round dining table. The cover of radius $r = 28$ cm has six identical floral border designs along its perimeter, leaving a central regular hexagonal region.

(i) Find the central angle subtended by each of the six design segments at the centre of the table cover. [1]

(ii) Calculate the area of the equilateral triangle formed by connecting the centre to the endpoints of one design. [1]

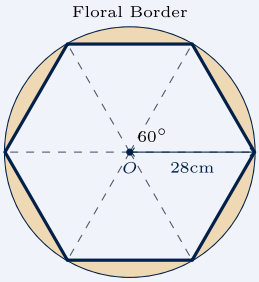
(iii) Calculate the total area of all six embroidered designs. $\left(\text{Use } \sqrt{3} = 1.732, \pi = \frac{22}{7}\right)$ [2]

— OR —

Find the total cost of embroidering the designs at the rate of Rs. 0.50 per cm^2. [2]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given: Radius $r = 28$ cm, Number of identical designs $n = 6$.



Part (i): Central Angle of Each Design Segment [1 Mark]

The full circle subtends 360° at the centre:

$$\theta = \frac{360^\circ}{6} = 60^\circ$$

Part (ii): Area of One Equilateral Triangle [1 Mark]

Since $\theta = 60^\circ$ and two sides are radii (28 cm), the triangle is equilateral with side $a = 28$ cm.

$$\text{Area of equilateral } \triangle = \frac{\sqrt{3}}{4}r^2 = \frac{\sqrt{3}}{4} \times (28)^2 = \frac{\sqrt{3}}{4} \times 784 = 196\sqrt{3} \text{ cm}^2$$

$$= 196 \times 1.732 = 339.472 \text{ cm}^2 \approx 339.47 \text{ cm}^2$$

Part (iii): Total Area of All Six Designs [2 Marks]

$$\text{Area of one sector} = \frac{60^\circ}{360^\circ} \times \pi r^2 = \frac{1}{6} \times \frac{22}{7} \times 28 \times 28 = \frac{1232}{3} \text{ cm}^2$$

$$\text{Area of 1 design segment} = \text{Area of sector} - \text{Area of } \triangle = \frac{1232}{3} - 196\sqrt{3}$$

$$\text{Total Area of 6 Designs} = 6 \times \left(\frac{1232}{3} - 196\sqrt{3}\right) = 2464 - 1176\sqrt{3}$$

$$= 2464 - (1176 \times 1.732) = 2464 - 2036.832 = 427.168 \text{ cm}^2 \approx 427.17 \text{ cm}^2$$

— OR (Alternative Choice) —

Total Cost of Embroidering the Designs: [2 Marks]

$$\text{Total Cost} = \text{Total Area} \times \text{Rate} = 427.168 \times 0.50 = \text{Rs. } 213.584 \approx \text{Rs. } 213.58$$

(i) $\theta = \mathbf{60^\circ}$, (ii) Area of $\triangle = \mathbf{339.47}$ cm^2, (iii) Total Design Area = **427.17** cm^2 (OR Cost = Rs. **213.58**

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